

Vishal Kumar

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EDUCATION

Indian Institute of Technology, Kanpur BS in Physics <i>Minor: Application of ML</i>	2022 – Present
Vishal Bharti Public School, Delhi Class XII (CBSE) <i>Stream: Physics, Chemistry, Mathematics (PCM)</i>	2021
Jawahar Navodaya Vidyalaya Sheikhpura Class X (CBSE)	2019

EXPERIENCE

Machine Learning Intern Optimspace	May 2025 – July 2025
- Fine-tuned YOLOv5 on 15k images, boosting mAP@0.5 by 18% via data augmentation for motion blur and occlusion - Integrated PaddleOCR with OpenCV preprocessing to maximize text recognition accuracy in low-contrast scenes - Optimized model inference for edge deployment by converting weights to TensorFlow Lite via post-training quantization, achieving 20 FPS real-time performance on mobile hardware with minimal accuracy loss - Implemented an automated continuous learning workflow to flag high-loss edge cases and periodically retrain the model, ensuring sustained detection robustness against glare and varying lighting conditions	

PROJECTS

End-to-End RAG Pipeline for Knowledge-Driven Text Generation <i>Self</i>	Aug – Sep 2025
- Architected a scalable RAG pipeline using LangChain and HuggingFace Transformers to enable context-aware inference on unstructured data. - Engineered a high-dimensional vector retrieval system using FAISS and Sentence-Transformers, utilizing Approximate Nearest Neighbor (ANN) search to optimize query latency. - Implemented semantic text chunking via spaCy/NLTK to maximize LLM context window utilization and minimize hallucination rates. - Designed a robust prompt engineering framework to enforce factual grounding, validating retrieval accuracy and domain adaptability.	
Autonomous Robot Navigation using DDQN <i>Deep Reinforcement Learning</i>	Jan-April 2026
- Developed DDQN reinforcement learning agent for autonomous robot navigation, achieving 88% loss reduction and 4-6x performance improvement over baseline across progressive difficulty levels - Trained neural network with experience replay and double Q-learning, reaching -490 mean reward and 25-30% success rate on blinking box environment (Level 2) - Designed 3-tier difficulty framework: Level 1 (static: 30-35% success), Level 2 (blinking: 88% loss improvement), Level 3 (moving-blinking: 800 episodes, validated convergence) - Deployed production-grade evaluation pipeline with reproducible seeding, multi-run statistics, and wall-obstacle configurations; pre-trained weights: 80.6-419.5 KB across difficulty levels	
Breaking ML-PUFs with Linear Models <i>Introduction to ML (CS771)</i>	Mar–April 2025
- Developed a C-SVM ML model by transforming 8-bit challenge vectors into highly informative 65-dim feature vectors - Validated using multiple linear classifiers, achieving 99.31% accuracy on the test set with a Logistic Regression model - Optimized the mapping function's time complexity while reducing training time by 31% and mapping time by 37% - Implemented optimized hyperparameters by balancing accuracy & training time, using C & tol plots for trade-off analysis	

SKILLS

Languages: Python, C++, C, MySQL, Bash
Libraries & Frameworks: TensorFlow, PyTorch, Keras, Scikit-learn, OpenCV, Transformers, LangChain, ChromaDB, NLTK, Pandas, NumPy, Matplotlib, Seaborn, Streamlit
Developer Tools: Git, GitHub, Jupyter, Google Colab, LaTeX

LEADERSHIP

Senior Executive Google Developer Student Clubs (GDSC)	IIT Kanpur
Established the community as a founding member and spearheaded hackathons in AI/ML, driving participation from 1500+ students.	
Captain Institute Cricket Team	IIT Kanpur
Led the squad to secure Gold in the General Championship and represented IIT Kanpur in national-level tournaments.	
Event Organizer Prabandhan	IIT Kanpur
Led logistics and execution for the inaugural Prabandhan (DoMS flagship fest), coordinating cross-functional teams to engage 6000+ students and set the foundation for future editions.	